General Science

Alveary

General Science: Grade 4

GRADE(S): 4

About the Course

Learners engage with microscopes, airplanes, and sound as they practice beginning lab skills and learn to construct bar graphs from their data. The lab component is required for this course. Coordinating afternoon activities are provided in Outdoor Work.

This topic is included in the following course(s): Science: Grade 4

About Science: Grade 4
In level 4, learners begin to notice science in their community and begin to develop their own interests as they discover a variety of topics from the microscopic world, physical science, and more. A mixture of more and less challenging books is included in this course.

Science: Grade 4
The Big Picture:
To accomplish the goal of supporting a relationship with the Things of the Universe, a Mason science program consists of nature lore, natural history, and general science. Nature immersion, inquiry, community connection, and supportive literature are woven into each of these three parts. Building on the familiarity and foundational knowledge of Form 1, Form 2 learners in grades 4-6 extend the scope of the Things they encounter and begin to notice that science is active in their world. Growing in their self-knowledge at this age, they may notice that they gravitate toward some of these Things more than others. This is a bit like noticing that you have a natural rapport with a particular friend because they seem to understand the way you think and share the same interests. This is wonderful and very typical, but we still want them to get to know more "friends" in science and try different Things, even though one might be their particular interest. Their sense of community is expanding, as well: who are the scientists in their world, and what are they doing? Special care has been taken to build vocabulary and conceptual knowledge "by the way" and to maintain the connection between scientific endeavors and the human experience. In Form 3, students will continue to build on this knowledge of what science is doing in the world when they examine science in its historical context.

Nature lore is timeless knowledge that is passed through a community, much like a grandmother passes on how to make that special bread when the dough just "feels right." Like Mason, we strive to pass on this knowledge primarily through outdoor work. Group nature walks, seasonal readings, and topics in scouting are provided as an Outdoor Work resource in the Quick Links. If desired, literature suggestions to support lore can be found in the Community Read Alouds resource (in Citizenship Grades 4+).

Placement & Combining Tips

About Science: Grade 4
In level 4, learners begin to notice science in their community and begin to develop their own interests as they discover a variety of topics from the microscopic world, physical science, and more. A mixture of more and less challenging books is included in this course.

General Science: Grade 4
For fourth-grade students or possibly hungry third-graders or fifth-graders taking their time. If combining grades is desirable: the combined grades may begin at the appropriate level and move forward together, or teachers can adapt the reading level and laboratory content to meet their needs.

Scheduling

GRADE	SCHEDULE INFO.	BOOKS
4	General Science: Grade 4 1 time/week 20 min	The Elephant Scientist To Fly: The Story of the Wright Brothers The Universe in You: A Microscopic Journey

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Science: Grade 4				
Natural History: Grade 4		General Science: Grade 4 Nature Walks & Scouting: Grades 1-8		Labs: Grade 4 Nature Notebook: Grade 4

Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Science: Grade 4
☐ Obtain any supplies indicated on the science or grade-level supply lists.
☐ Download any apps and shortcut any desired links.

Nature Lore:
☐ Bookmark your Outdoor Work Quick Link, so that you have it available on your weekly outing. Outdoor Work is generally flexible for your location and season and can be moved around in the schedule to incorporate or substitute Natural History Club outings.
☐ Print or bookmark grade-specific nature notebook suggestions to support natural history and general science. Notebooking can be done on a walk, during occupations, or as a field trip, as appropriate.

Special Topics & Field Trips

Science: Grade 4
Encourage students to begin choosing their own special topic to observe, study, and research in their field guides. What or who are they curious about or interested in getting to know better? Teachers can choose their own special study, too! Suggested field trips include:

General Science: Grade 4
☐ Term 1: a museum with a microscopy exhibit
☐ Term 2: an airport observation deck and/or a museum with airplanes
☐ Term 3: a museum with a sound exhibit and a zoo

Term Prep & Teacher Tips

Science: Grade 4
Object lesson prompts are woven into the lessons, as appropriate, but teachers are encouraged to read corresponding Handbook of Nature Study (HoNS) selections each term and/or any that support special topics chosen by students:

General Science: Grade 4
☐ Term 1: introduction to rocks p.743-745
☐ Term 2: one or more birds from p.62-142
☐ Term 3: sedimentary, igneous, and metamorphic rocks p.745-748

☐ Gather household items, typically easy for students to scavenge or teachers to obtain locally:

- lighter or a match for candle
- penny
- flashlight
- a few salt crystals
- water
- table or counter with overhang
- hair dryer
- a few sheets of multipurpose paper
- long hallway for flying paper airplanes
- large-medium-sized bowl
- paper towels
- empty can or cup
- voice recorder, such as found on most electronic devices
- friendly animal, such as yours or a neighbor's pet (Term 3)
- printables
- optional: computer
- optional: sewing needle
- optional: metal coathanger
- optional: metal spoon
- optional: 2 pieces of string ~30cm

Reminders

General Science: Grade 4

☐ Note that there is an activity during Lesson 1. Be sure to have your materials ready!

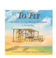
Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ [View Book List Details]

General Science: Grade 4


The Elephant Scientist


To Fly: The Story of the Wright Brothers


The Universe in You: A Microscopic Journey

Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ [View Basic Supplies]

∞ [View Supply List Details]

General Science: Grade 4


Hand Lens

Household Items - General Science: Grade 4

Quick Links

Click THIS text or scan the QR code for links.

Related Course Quick Links

∞ [Extra Helpings]

∞ [Outdoor Work]

- Microscope Coloring Book
- Seek app from iNaturalist
- SkyView Lite for iOS
- SkyView Lite for Android
- Foundations (See Section 13: Science)

General Science: Grade 4

How To Teach

Introduce

• Begin each lesson with learners' existing knowledge. If the book or activity is new or unfamiliar, then look at the title, a picture, or guidance in the lesson plan to help discuss what students think, drawing on previous experience. If continuing or revisiting a topic or activity, then recap.

• Often, the introduction in the lesson plans or the title of the book's section can help learners to draw out the main idea. Use this to support learners, as appropriate.

• Some learners may benefit from using pictures or looking back briefly.

• Allow them time to share any concerns and come alongside, as needed.

Read

• Read or do, as instructed in the lessons, noting any Teacher Tips provided. Learners should always have their journal, notebook, or other paper available in case they need to draw or diagram during the lesson.

• Use supportive strategies and educational tools to reduce frustration and better engage the mind, as appropriate. These could include, but are not limited to, the use of eBooks, pictures, audio, read-aloud, buddy reading, colored reading strips, etc.

• If learners do not understand a word or concept, do not worry. Try to show them with a picture or connect the idea to something they have seen in real life. They are learning much by the way and will likely build understanding over the term, the year, and beyond.

Narrate

• Process the ideas of the lesson by retelling events, describing, explaining a concept, etc. Tips about helping learners deal with various types of passages are provided, but teachers can learn more in Charlotte Mason's School Education Chapter 16, especially p.180.

• Learners may use words, pictures, Legos, PowerPoint, etc, to process and convey ideas in their own way.

• Teachers may take turns to model.

Discuss

• Consider together any thoughts, confusion, or concerns about the passage, keeping in mind that oftentimes new concepts are not going to be 'mastered' on the first or even second introduction. Mastery is not necessarily the goal, but curiosity and thinking.

• Questions/topics for further discussion are often provided in the lesson plans (or even lab books) to help. There are no right or wrong answers to these. Alternatively, many of these can be used for composition, depending on the needs of the learner and the instructional goals of the teacher. If teachers want to keep ideas active in the mind, these can also be used at other times to keep the ideas in the working memory.

• Note that learners may need to spend more than the allotted time engaging with or even repeating a lesson before moving on or as reinforcement at a later time. Adjust the pace as needed to feed the learner.

• Notice if there were any dates that they want to keep for their Book of Centuries.

Connect

- Follow any extra links, examine any sidebars in the text, look at pictures, etc., depending on learner needs and interest.

- These can be viewed as alternative ways to engage.

General Science: Grade 4

[Click THIS text or scan the QR code for links.]

Term 1

WEEK 1

⬚ **20m General Science: Grade 4 - Lesson 1**

Zooming In

> ⎕ Materials: The Universe in You, penny/dime, ruler, hand lens, image link

➔ INTRO
What's the smallest Thing you've ever seen? Do you think there is anything smaller? Look at the picture on the cover. What do you think you are seeing? Did you know that there are such Things that are so small that we can't usually see them? This book is about the smallest Things we know of. You can use your ruler to track how small Things are for the first few pages.

➔ READ
p.1-8 "The Calliope Hummingbird" - "of a vellus hair."

➔ ACTIVITY
Can you see your skin cells or bacteria? How can we know they exist? Sometimes we can sense the effect of something we can't see, like the wind; sometimes we can use a tool to make it visible. To look closer at small Things, scientists use a tool called a microscope. We do 2 things with a microscope: magnify and focus.
- Look at ∞ Image Link: Penny & Salt or Dime & Salt
- This image has been magnified. Compare it to your coin. What does it mean to be magnified? How much bigger is it? How many of your coins could fit across this image?
- Did you notice that there are some salt crystals on the magnified coin? How much has the salt been magnified? (same as the coin)
- Use your hand lens to look at your coin. What details do you notice that you couldn't see without the hand lens? Look at the other side:

Penny: Can you find the little man between the columns of the building? Notice what happens as you move the penny and lens closer together/farther apart. What does it mean to focus?
Dime: Notice that the plants on either side of the torch are different. One is an olive and the other an oak. Can you tell the difference? Notice what happens as you move the dime and lens closer together/farther apart. What does it mean to focus?

➔ NARRATE & DISCUSS

★ TEACHER TIP
Spend as much time as is wanted looking at the artwork in this book before going on to each successive page. Share the captioned facts, as desired.

★ TEACHER NOTE
Note that next week is an activity day. Check your supplies!

WEEK 2

⬚ **20m General Science: Grade 4 - Lesson 2**

The Microscope

> ⎕ Materials: video/image links, microscope, flashlight, slide, salt crystals

➔ INTRO
What did we learn about in the last lesson? How do we see very small Things? What two things does a microscope do? Let's look at a very old microscope, and then we'll compare that to ours. Watch for him to show you the tube where the magnifying lenses are and the mirrors that direct the light.

➔ VIEW
∞ Video Link: A 1750s Compound Microscope (5:06)

➔ ACTIVITY
- Where is the tube and where are the lenses on your microscope? Refer to the diagram that came with your microscope. The one at the link may help for comparison. How much does the lens at your eyepiece (the ocular lens) magnify? How much do the lenses at the bottom of your tube (the

objective lenses) magnify?
∞ Image Link: Compound Microscope Parts
• How did he use the mirror? Where is your light source? Place a slide with a few salt crystals on the stage. Does this direct light through the specimen or onto the surface? Look at the crystals and describe/draw what you see.
• Turn the light source off and instead shine the flashlight down onto the surface of the salt crystals. Adjust the flashlight to point as much light onto the surface as you can. Look at the crystals and describe how this is different from the lower light source.

→ NARRATE & DISCUSS

WEEK 3

20m General Science: Grade 4 - Lesson 3
Zooming in Further

Materials: The Universe in You

PREP: Read Teacher Tip

→ INTRO
What did the girl in this book see last time? Look at the pictures, as needed.

→ READ, NARRATE, & DISCUSS
p.9-end "The smallest hairs" - "the universe within."

★ TEACHER TIP
Students don't need to remember all of the scientific language. Just let them hear it "by the way."

WEEK 4

20m General Science: Grade 4 - Lesson 4
Imagining

Materials: poem link, image links

PREP: Read Teacher Tip

→ INTRO
We've been learning about very small Things. Tell me about Things too small to see. How can we know they are there? Some of the Things in your book (and in the lab) are so small that you won't be able to see them even with your microscope. Sometimes, when we first begin looking very closely and using a microscope, it can be hard to know what we are looking at. Read our poem for today. What do you think the author would say about the strange images we see under the microscope?

→ READ, DISCUSS, & REREAD
∞ Website Link: The Microbe

→ ACTIVITY
Even while we keep looking and asking questions, imagining what the image reminds us of can help us think about what we're seeing. Check out the imaginings at the link, and then draw or describe some of your own using the images in the Science Photo Gallery or some that you have seen under your microscope.
∞ Image Link: Microscope Imaginings
∞ Image Link: Science Photo Gallery

★ TEACHER TIP
Students who find copywork/dictation challenging may be anxious or unsure about today's activity. Encourage them to dictate, work in tandem, or even try a digital drawing program, if this is available.

• DEFINITIONS
sanguine: marked by eager hopefulness : confidently optimistic

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WEEK 5

20m General Science: Grade 4 - Lesson 5
Louis Pasteur

Materials: video link

PREP: Read Teacher Tip

→ INTRO

★ TEACHER TIP
Are your learner(s) demonstrating more ease in using the microscope? More interest/ease in describing what they see with it or how it works? If not, consider practicing together more often, exploring extra helpings, or taking

Term 1

Tell me what we were thinking about in the last lesson. The invention of the microscope allowed many scientists to begin looking closer and asking questions. Louis Pasteur is one of the most famous. Let's hear his story today.

→ VIEW, NARRATE, & DISCUSS
∞ Video Link: Louis Pasteur (10:36)

● Pasteur said, "Do not let yourself be tainted with barren skepticism." What does that mean? How do we ask questions, like the poet who wrote the Microbe suggests, without being "tainted with barren skepticism"?

a field trip to a museum or lab with microscopes. Reach out through Contact Us if you need help.

WEEK 6

⬚ **20m General Science: Grade 4 - Lesson 6**
Microscopy Questions Today

Materials: video link

→ INTRO
What have we been learning about? Who was Louis Pasteur, and what kinds of questions did he ask? What kinds of questions do you think people ask with their microscopes today? For the next few weeks, we'll learn about some of today's questions.

→ VIEW, NARRATE, & DISCUSS
∞ Video Link: We Don't Know Why Moth Wings Glow (10:09)

★ TEACHER NOTE
The lesson video mentions evolution and adaptation during 5:37-6:40. Teachers might choose to introduce the terms by the way, forward through that portion, or watch without some or all of the narration.

WEEK 7

⬚ **20m General Science: Grade 4 - Lesson 7**
Microscopy Questions Today

Materials: video link

→ INTRO
What did we see last week? Today we have another question that scientists are using microscopes to investigate.

→ VIEW, NARRATE, & DISCUSS
∞ Video Link: The Cryptic Origins of Yogurt (10:21)

WEEK 8

⬚ **20m General Science: Grade 4 - Lesson 8**
Microscopy Questions Today

Materials: video link

→ INTRO
Tell me about our video from last week. Let's learn about one more question today.

→ VIEW, NARRATE, & DISCUSS
∞ Video Link: Tiny Mysteries from the Black Sea (8:07)

WEEK 9

⬚ **20m General Science: Grade 4 - Lesson 9**
The Universe in You

Materials: The Universe in You

→ INTRO
Tell me about some of the questions that scientists are using microscopes to investigate. Now that you've seen your own cells in the lab, let's read our book one more time.

→ READ, NARRATE, & DISCUSS
The Universe in You (whole book)

General Science: Grade 4

[Click THIS text or scan the QR code for links.]

Term 1

WEEK 10

⬚ **20m General Science: Grade 4 - Lesson 10**
Zooming In Even More

▯ Materials: website link, printable

PREP: Read Teacher Tip

→ INTRO
What are some of the most interesting Things you've looked at with your microscope? What else would you like to see? You may have noticed that when you zoom in as much as you can, the image starts to get blurry. Once they zoom in that far, scientists have to use another kind of microscope to zoom in even more. The Science Photo link contains a gallery of images using these special microscopes. The scientist who prepared these images added color to help draw our attention to certain features. Explore the gallery.

→ VIEW
∞ Website Link: Science Photo Gallery

→ ACTIVITY
The links below contain a collection of original images without color that a scientist shared especially for you. You may choose images from this collection to add your own color. Which features do you find most interesting? Which ones do you want your viewer to notice?
∞ Printable Link: Electron Microscope Coloring Pages

★ TEACHER TIP
Students who find copywork/dictation challenging may be anxious or unsure about today's coloring activity. Encourage them to dictate, work in tandem, or even try a digital coloring program, if this is available.

WEEK 11

⬚ **20m General Science: Grade 4 - Lesson 11**
Catch Up

PREP: Read Teacher Tip

→ CATCH UP
Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

★ TEACHER TIP
When student(s) take exams next week, evaluate what they recall, but more importantly, whether the microscope, people, or Things they have seen are more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less business, etc? Reach out through Contact Us if you need help.

WEEK 12

⬚ **20m General Science: Grade 4 - Lesson 12**
Exam Week

→ EXAMS
Answer question(s) related to course.

Questions will come from:
The Universe in You and multimedia content

General Science: Grade 4

Click THIS text or scan the QR code for links.

Term 2

WEEK 1

⬚ **20m General Science: Grade 4 - Lesson 13**

Inspiration

▯ Materials: To Fly

PREP: Read Teacher Tip

→ INTRO
Look at the picture on the cover. Do you know who the Wright Brothers are? What do you know about them? Read and discuss the poem called "Crazy Boys" in the early pages of the book.

→ READ, NARRATE, & DISCUSS
p.7 "People had always" - "For a time."

● How do you think the toys that inspired the Wrights worked? Can you tell from the picture? Compare to this image from the Smithsonian Museum:
∞ Image Link: Smithsonian: Orville's Sketch of Penaud's Helicopter

→ SUPPLEMENTAL
Notice how many times the craftsman in the video fails before he gets it right, but how wonderfully it works when he does.
∞ Video Link: YouTube: Making a Penaud Helicopter (5:04)

★ TEACHER TIP
As you read, notice the purpose of the passage. Then, if students struggle to narrate, you can prompt them by helping them identify the purpose. Is it describing a scene or a person? Telling a sequence of events? Explaining an idea?

★ TEACHER TIP
Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to the discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

WEEK 2

⬚ **20m General Science: Grade 4 - Lesson 14**

Observing Motion

▯ Materials: To Fly

PREP: Read Teacher Tip

→ INTRO
Recall how the story began. Do you think the propeller on their toy helicopter is a simple machine? If so, which might it be?

→ READ, NARRATE, & DISCUSS
p.8 "Wilbur and Orville" - "than flat ones."

● Do you think the kites that Orville built were simple machines? If so, which one? If not, why not? Spend any remaining lesson time noticing objects or tools that make work and motion easier. This is a good indication that there is probably a simple machine at work.

★ TEACHER TIP
Hinges, knives, jar lids, any piece of playground equipment, the pulley on a flagpole, garage doors, stairs, scissors, etc., are all simple machines. Help students notice that they are everywhere while discussing.

WEEK 3

⬚ **20m General Science: Grade 4 - Lesson 15**

Learning Together

▯ Materials: To Fly

→ INTRO
Tell me about the Wrights so far. Let's find out what happens when they grow up.

→ READ, NARRATE, & DISCUSS
p.10-13 "In time, Orville" - "the right answer."

● Disagreeing and arguing seemed to help Orville and Wilbur. Do you feel the same way when someone disagrees with you? If so, how does it help you? If not, why not? When you disagree with someone, do you feel like you are trying to help them?

General Science: Grade 4

[Click THIS text or scan the QR code for links.]

Term 2

WEEK 4

20m General Science: Grade 4 - Lesson 16

Kites Become Gliders

Materials: To Fly

→ INTRO
Where did we leave the Wrights? Recall what happened. How do you think they got from bicycles to airplanes?

→ READ, NARRATE, & DISCUSS
p.15-18 "In 1896, Orville" - "was build it."

● Describe the 3 movements described in the passage. You can tell, draw, or use your body to do this. Show the special mechanism that would allow the Wrights to control the movements.

WEEK 5

20m General Science: Grade 4 - Lesson 17

Experiments

Materials: To Fly

PREP: Read Teacher Tip

→ INTRO
Recall what happened in the last passage.

→ READ, NARRATE, & DISCUSS
p.19-22 "Wilbur wrote to" - "problems of flight."

● There were many highs and lows during the Wrights' engineering process. Have you ever worked at something that got you both excited and frustrated? Did you want to give up?

★ TEACHER TIP
Are your learner(s) demonstrating more ease in exploring how Things work? More interest/ease in thinking about the Wright Brothers or their work? If not, consider playing with mechanical toys together, exploring extra helpings, or taking a field trip to a park to observe the equipment. Reach out through Contact Us if you need help.

WEEK 6

20m General Science: Grade 4 - Lesson 18

Using the Right Information

Materials: To Fly

→ INTRO
Recall what happened in the last passage.

→ READ, NARRATE, & DISCUSS
p.24-27 "The brothers began" - "self-propelled flying machine."

● What do you think helped the Wrights to persevere and keep going even though they had been so frustrated in the last reading?

WEEK 7

20m General Science: Grade 4 - Lesson 19

Designing a Flyer

Materials: To Fly

→ INTRO
Recall what happened in the last passage.

→ READ, NARRATE, & DISCUSS
p.28-32 "During the winter" - "a homemade engine?"

General Science: Grade 4

[Click THIS text or scan the QR code for links.]

Term 2

WEEK 8

⬚ **20m General Science: Grade 4 - Lesson 20**
The First Attempt

▯ Materials: To Fly

PREP: Read Teacher TIp

➔ INTRO
Recall what happened in the last passage.

➔ READ, NARRATE, & DISCUSS
p.33-34 "On Monday, December" - "a successful flight."

➔ SUPPLEMENTAL
Examine the photographs in the link, especially photos 1, 2, and 5. Notice what the differences were between their gliders and the airplane. If interested, draw or diagram any or all of them, showing the important features.
∞ Image Link: Wright Brothers Memorial

★ TEACHER TIP
Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to the discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

WEEK 9

⬚ **20m General Science: Grade 4 - Lesson 21**
The First Flight

▯ Materials: To Fly

➔ INTRO
Recall what happened in the last passage. Do you think they will try again?

➔ READ, NARRATE, & DISCUSS
p.36-41 "It took two" - "Aviation had begun."

• Their friend said they seemed like two people "who weren't sure they'd ever see each other again." What does it say about their characters that they kept trying even though they knew they could die? Why would they do this?

➔ SUPPLEMENTAL
∞ Image Link: Library of Congress: First Flight

• NATURE NOTEBOOK
Prompt for General Science: Grade 4 in Outdoor Work Quick Link.

WEEK 10

⬚ **20m General Science: Grade 4 - Lesson 22**
Lessons from their Story

▯ Materials: To Fly

➔ INTRO
Recall what happened in the last passage. What do you think the world thought about this news?

➔ READ, NARRATE, & DISCUSS
p.42-44 "The brothers packed" - "actually did it."

• Everyone has strengths and challenges, even the Wright Brothers. Talk about how you have noticed this in yourself or your friends and family.

• NATURE NOTEBOOK
Prompt for General Science: Grade 4 in Outdoor Work Quick Link.

WEEK 11

⬚ **20m General Science: Grade 4 - Lesson 23**
Catch Up

PREP: Read Teacher Tip

★ TEACHER TIP
When student(s) take exams next week, evaluate what they recall, but more importantly, whether the airplane, the Wright Brothers, any mechanical Things, or any Things that they have met are

General Science: Grade 4

[Click THIS text or scan the QR code for links.]

Term 2

	➔ CATCH UP Use this time to catch up on lessons in this or another course, or explore Extra Helpings.	more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less business, etc? Reach out through Contact Us if you need help. **• FIELD TRIP** Visit a location that has airplanes. The Wright Brothers National Memorial, an Air and Space museum, or an airport observation deck are examples.
WEEK 12	⬚ **20m General Science: Grade 4 - Lesson 24** *Exam Week* **➔ EXAMS** Answer question(s) related to course. Questions will come from: To Fly	

General Science: Grade 4

[Click THIS text or scan the QR code for links.]

Term 3

WEEK 1

⬚ **20m General Science: Grade 4 - Lesson 25**
It Reminds Me Of

▯ Materials: The Elephant Scientist, globe

➜ INTRO
Look at the picture on the cover and the title. Have you seen elephants at a zoo? What do you remember about them? Do you think they communicate with each other? If so, how? This is a story about scientists who are trying to explain how they communicate and what they are saying. Most of the elephants in this book live at Etosha National Park and Mushara Waterhole. You can find it on the map on the page just before the Table of Contents. Can you find the location on your globe?

➜ READ, NARRATE, & DISCUSS
p.1-3 "Caitlin O'Connell peered" - "sense their environment."

WEEK 2

⬚ **20m General Science: Grade 4 - Lesson 26**
Becoming a Scientist

▯ Materials: The Elephant Scientist

PREP: Read Teacher Tip

➜ INTRO
Recall what happened in the last passage.

➜ READ & NARRATE
p.4-6 "Caitlin O'Connell studies" - "her animal studies."

➜ READ, NARRATE, & DISCUSS
p.7-10 "After working with" - "she never anticipated."

● Do you think it would be challenging to study and learn about something that you cannot see? Why or why not? How can technology, like you see in the link, help scientists study the invisible?
∞ Video Link: What does Sound look like? (2:32)

★ TEACHER TIP
Some students may be able to read the entire passage and narrate the scientist's journey, but others may benefit from having it broken into smaller portions, as it is presented here. Do whatever is most appropriate for the student(s).

WEEK 3

⬚ **20m General Science: Grade 4 - Lesson 27**
To Be an Elephant

▯ Materials: The Elephant Scientist

PREP: Read Teacher Tip

➜ INTRO
Recall what happened in the last passage.

➜ READ, NARRATE, & DISCUSS
p.11-16 "African elephants reign" - "consume enough food."

➜ SUPPLEMENTAL
∞ Video Link: Learning to be an Elephant (1:51)

➜ NOTE
Read "Elephant Species" on p.17, if desired.

★ TEACHER TIP
Supplemental links for optional support are provided at the END of the lesson. They can be used at any point: to help generate interest during the introduction, to enliven the lesson itself, or to add to the discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

General Science: Grade 4

[Click THIS text or scan the QR code for links.]

Term 3

WEEK 4

⬚ **20m General Science: Grade 4 - Lesson 28**
To Understand an Elephant

Materials: The Elephant Scientist

PREP: Read Teacher Tip

→ **INTRO**
Recall what happened in the last passage.

→ **READ & NARRATE**
p.18-19 "Elephants may be" - "in the field."

→ **READ, NARRATE, & DISCUSS**
p.20-23 "Caitlin and Tim" - "from the area."

● Do you think different parts of your brain help you do different things, like the elephants? Find the temporal lobe in the diagram at the link. Recall what the temporal lobe did for the elephant. Where is your temporal lobe? (Point to it.) Try humming different music that you know. Try a happy song, a sad song, a scary song. Your temporal lobe helps you think about and remember those tunes and how they make you feel!
∞ Image Link: Mayo Clinic: Brain Lobes

★ TEACHER TIP
Again, this reading is presented in smaller portions because this strategy is helpful to some students. Others may be able to read the entire passage before narrating. Do whatever is most appropriate for the student(s).

● **FIELD TRIP**
Visit a science museum that has a sound exhibit.

WEEK 5

⬚ **20m General Science: Grade 4 - Lesson 29**
Elephant Families

Materials: The Elephant Scientist

PREP: Read Teacher Tip

→ **INTRO**
Recall what happened in the last passage.

→ **READ, NARRATE, & DISCUSS**
p.24-28 "After the hectic" - "their unruly behaviors."

● Compare elephant families to human families. How are they similar, and how are they different?

→ **SUPPLEMENTAL**
∞ Video Link: Elephant Families (2:20)
∞ Video Link: Etosha National Park (3:42)

★ TEACHER TIP
Are your learner(s) demonstrating more ease in thinking about waves or sound? More interest/ease in elephants or animal communication? If not, consider observing together more often, exploring extra helpings, or taking a field trip to a zoo or museum with a sound exhibit. Reach out through Contact Us if you need help.

WEEK 6

⬚ **20m General Science: Grade 4 - Lesson 30**
The Hypothesis

Materials: The Elephant Scientist

→ **INTRO**
Recall what happened in the last passage.

→ **READ & NARRATE**
p.29-33 "It was during" - "alarm call playback."

→ **READ, NARRATE, & DISCUSS**
p.34-35 "When the dry" - "high-frequency components."

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Term 3

WEEK 7 ⬚ **20m General Science: Grade 4 - Lesson 31**

Settling into Camp

Materials: The Elephant Scientist

→ INTRO
Recall what happened in the last passage.

→ READ, NARRATE, & DISCUSS
p.36-41 "As in earlier" - "for the night."

WEEK 8 ⬚ **20m General Science: Grade 4 - Lesson 32**

The First Experiment

Materials: The Elephant Scientist

→ INTRO
Recall what happened in the last passage.

→ READ, NARRATE, & DISCUSS
p.42-44 "For the first" - "about impending danger."

WEEK 9 ⬚ **20m General Science: Grade 4 - Lesson 33**

More to Learn

Materials: The Elephant Scientist, optional: metal coathanger, metal spoon, 2 pieces of string ~30cm

→ INTRO
Recall what happened in the last passage.

→ READ & NARRATE
p.46-48 "Once Caitlin established" - "no olfactory cue."

→ READ, NARRATE, & DISCUSS
p.49-51 "As Caitlin continued" - "outright killing them."

● Act out the experiment Caitlin did to test how well Donna could sense vibrations.
● Try the demonstration suggested on p.51: "The easiest way to experience this ..." You can have fun with this by doing the same thing with a metal hanger as shown in the link. Replace the hanger with a metal spoon and see if the sound changes.
∞ Image Link: Coathanger Gong

★ TEACHER NOTE
There is a photograph from the dissection of an elephant's foot on p.50. Most students are not bothered when encountered by the way, and some will find it fascinating. However, if teachers are concerned, they might choose to cover the photo or give students the option to look or not.

WEEK 10 ⬚ **20m General Science: Grade 4 - Lesson 34**

More Questions

Materials: The Elephant Scientist

→ INTRO
Recall what happened in the last passage.

→ READ, NARRATE, & DISCUSS
p.52-59 "A long line" - "his left ear."

● Caitlin's work got her interested in questions about the bulls. How many times can you think of when Caitlin's work created new questions for her to pursue?
● Make up a poem about elephants together!

General Science: Grade 4

[Click THIS text or scan the QR code for links.](#)

Term 3

WEEK 11

⬚ **20m General Science: Grade 4 - Lesson 35**
More Work

▯ Materials: The Elephant Scientist

➔ INTRO
Recall what happened in the last passage.

➔ READ & NARRATE
p.60-61 "As the sun" - "a helping hand."

➔ READ, NARRATE, & DISCUSS
p.62-64 "Elephants once roamed" - "well they match."

➔ SUPPLEMENTAL
∞ Video Link: How Elephants Listen (4:33)

★ TEACHER TIP
When student(s) take exams next week, evaluate what they recall, but more importantly, whether sound, elephants, or animal communication have become more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less business, etc? Reach out through Contact Us if you need help.

● **FIELD TRIP**
Visit a zoo that cares for elephants. Read or review Elephant Species p.17 and notice which species is at your zoo.

WEEK 12

⬚ **20m General Science: Grade 4 - Lesson 36**
Exam Week

➔ EXAMS
Answer question(s) related to course.

Questions will come from:
The Elephant Scientist

General Science: Grade 4

Examination

Term 1

· The Universe in You: Tell about or draw what microscopes do. OR Tell about or draw something that you or another scientist has seen with microscopes.

· Share what you learned about using the microscope. Tell about a challenge in using this tool and how you worked to overcome this challenge.

Term 2

· To Fly: Tell about or draw how the Wright Brothers made their plane fly when no one else could. OR Tell how the Wright Brothers helped each other when they got frustrated.

· Explain how you did an investigation in the science lab this term and what you learned from it.

Term 3

· The Elephant Scientist: What are sound waves? OR Describe what you know about elephant communication.

· Explain how you did an investigation in the science lab this term and what you learned from it.